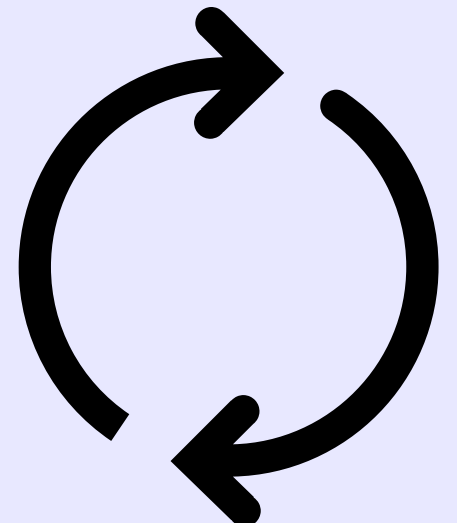
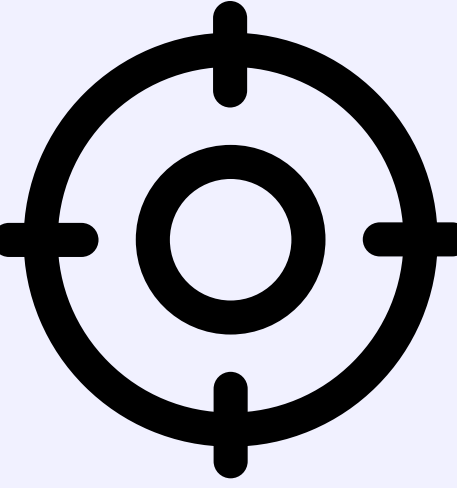
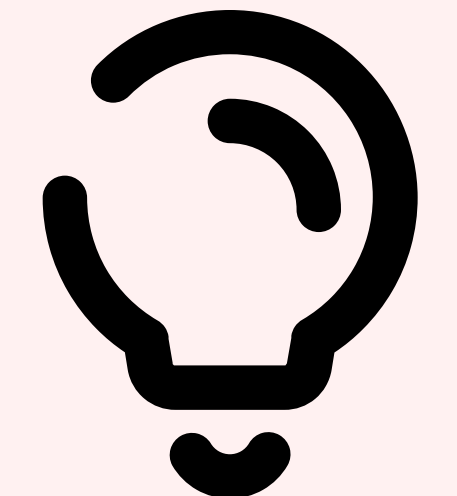
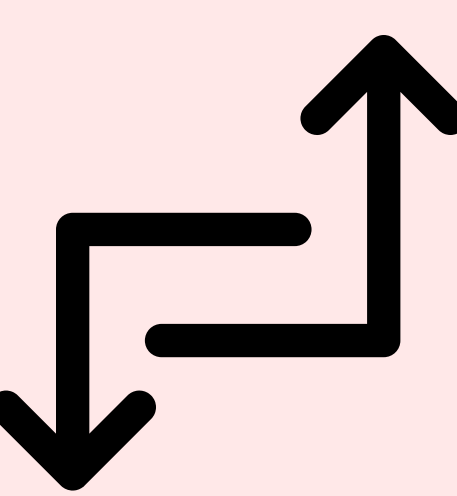


1-3. Diagnosis to Matched Intervention

143 failed SWE-bench Verified trajectories

Failure types ordered by scaffoldability (strong → weak)

Failure Type	Cascade Signature	Best Scaffold Δ vs. control (0-3 score)		Verdict
 EDIT n = 28 (20%)	Repetitive retry loop WR = 88.8%	reread_file	+ 1.00 ***	Scaffoldable
 PLAN n = 8 (6%)	Commitment spiral WR = 70.7%	step_back	+ 0.88 †	Scaffoldable
 LOC n = 37 (26%)	Mislocalized effort WR = 80.0%	reread_issue	+ 0.27	Marginal / risky
 LOGIC n = 70 (49%)	Reasoning drift WR = 81.2%	minimal_fix	+ 0.17	Frontier (reasoning-limited)



Mismatch harm: LOC + test_guided: 0.87 vs. 1.70 Generic retry Δ = -0.83

4. Type-Aware Routing

(Online Selection → Performance)

Policy performance

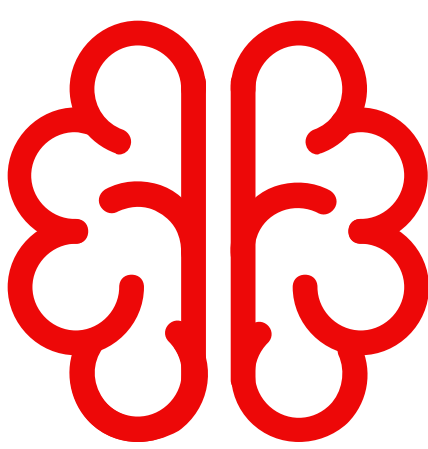
(mean score, 0-3) on evaluation set

Oracle type-specific
(upper bound) **2.32**

Predicted type
(online classifier) **2.17**

Universal reread_file
(fixed strategy) **2.04**

Generic retry
(control) **1.76**



Online classifier: 74.8% accuracy
Predicted-type routing captures
73% of oracle gain

Strategy pool: 10 pre-specified one-sentence scaffolds; prompts in Appendix.